DO NOW — RECALL

RAM, ROM and virtual storage

Name: _____ Class: _____ Date: _____

1. Why are SSDs and HDDs physically different?

2. What purpose does processor cache serve?

3. Which is normally volatile: RAM or ROM?

4. Sorted array [2,6,10,14,18,22,26,30]. Find 22 with midpoint indexes and range updates.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

SSDs use semiconductor cells; HDDs use rotating magnetic platters.

2. Model answer

Keeps some needed data/instructions in fast nearby memory.

3. Model answer

RAM is volatile.

4. Model answer

low0 high7 mid3→14; low4 high7 mid5→22, found at index 5.

Vocabulary — know these terms

RAM — Volatile read/write working memory.

ROM — Non-volatile firmware/boot storage.

virtual storage — Remote network-accessible file storage.

virtual memory — OS technique using address translation and backing storage.

firmware — Low-level code retained without power.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

Network outages, account security and mistakenly synced deletions are relevant risks.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

RAM versus ROM

RAM supports active execution and is volatile; ROM and firmware storage retain instructions without power.

Define virtual storage

A remote or cloud storage service keeps files on network-accessible systems, potentially available on multiple devices.

Distinguish virtual memory

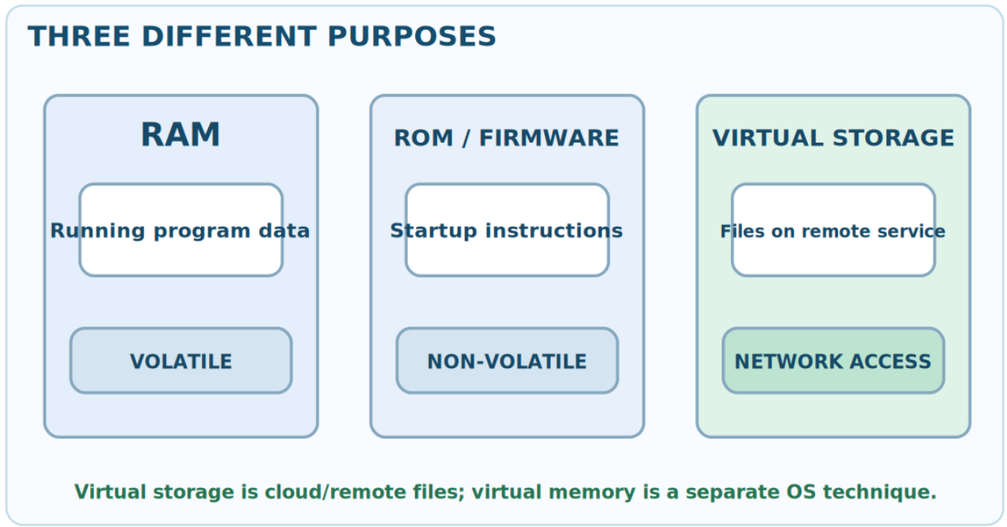
OS virtual memory involves memory addressing/paging with secondary storage, taught again in OS memory management.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

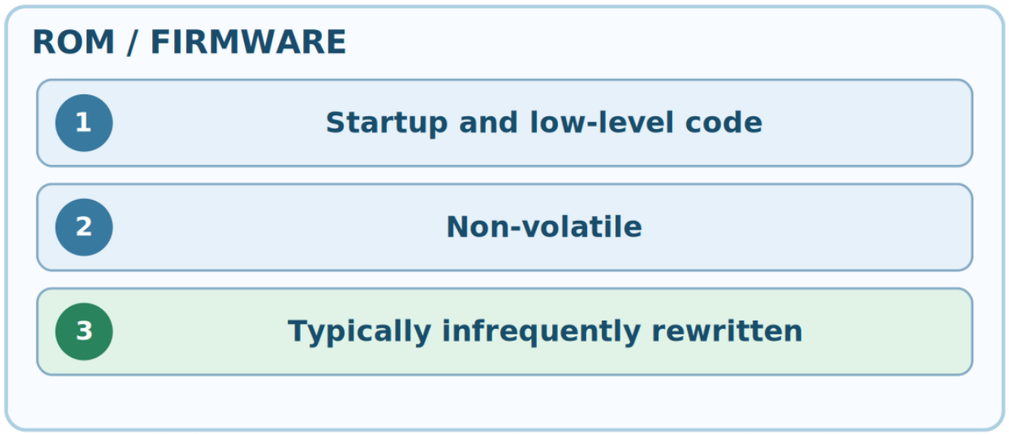
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



RAM volatile running data; ROM nonvolatile firmware; virtual storage remote network files

New Learning



ROM / FIRMWARE: Startup and low-level code; Non-volatile; Typically infrequently rewritten

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Virtual cloud storage is another name for OS virtual memory.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

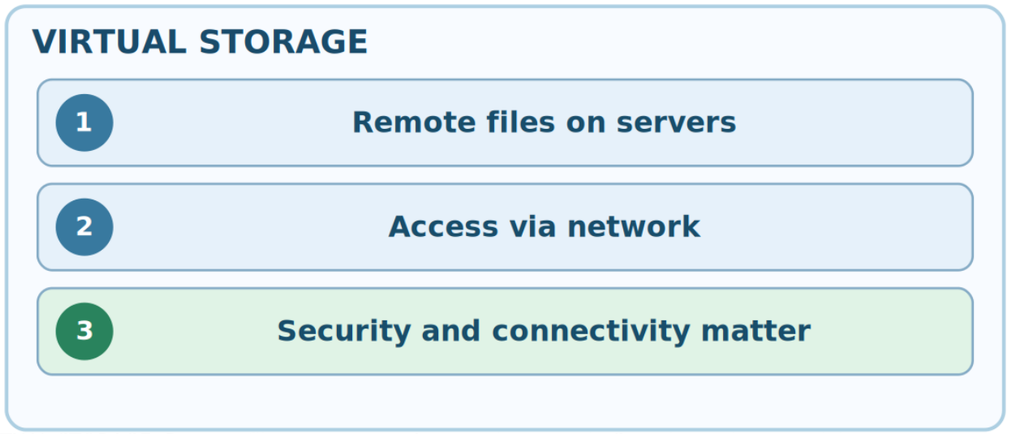
Question 3

Distinguish virtual storage from virtual memory and explain one limitation of each.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



VIRTUAL STORAGE: Remote files on servers; Access via network; Security and connectivity matter

I DO — WORKED EXAMPLE

Follow the worked method

Why can unsaved editing work disappear after power loss while firmware remains?

Predict

Unsaved temporary program state depends on volatile RAM and can disappear without power.

Explain

Firmware persists in non-volatile storage, allowing startup instructions to remain available.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State typical use and whether RAM and ROM are volatile.

Question 2

Explain two benefits and one drawback of network-based virtual storage for pupil coursework.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Virtual/cloud storage supports access from multiple locations and may offer remote copies.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

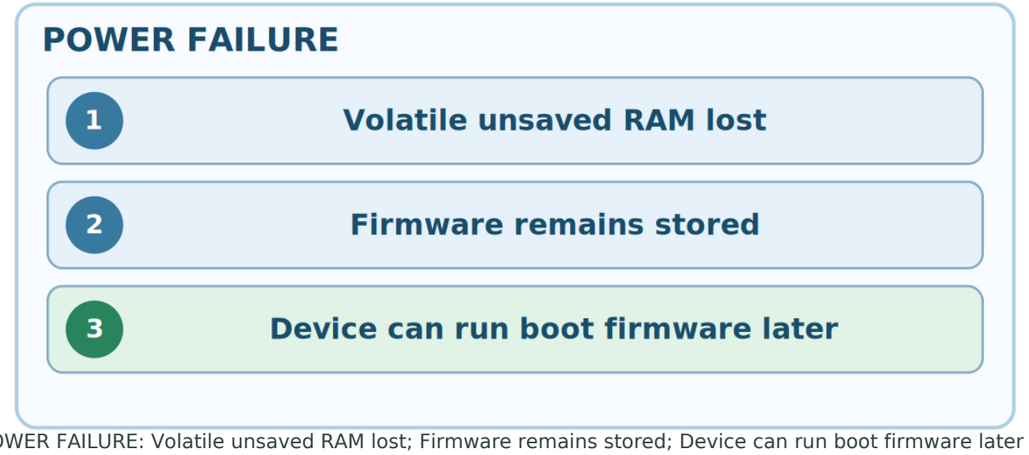
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Why can unsaved editing work disappear after power loss while firmware remains?

Unsaved temporary program state depends on volatile RAM and can disappear without power.



POWER FAILURE: Volatile unsaved RAM lost; Firmware remains stored; Device can run boot firmware later

WE DO — GUIDED PRACTICE

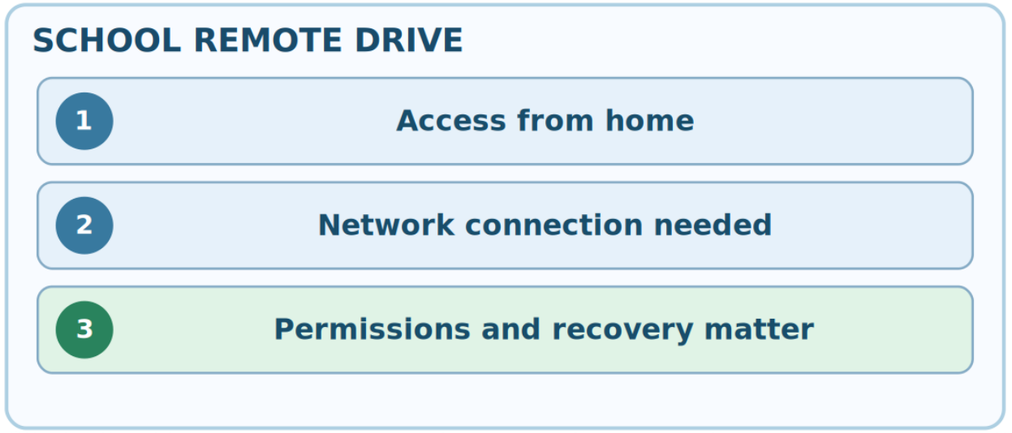
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil opens saved coursework from home using a school online drive. State advantages and risks.

Virtual/cloud storage supports access from multiple locations and may offer remote copies.



SCHOOL REMOTE DRIVE: Access from home; Network connection needed; Permissions and recovery matter